

School of Computer Science
COMP 2067 Programming for Beginners
Summer 2025 – Lab 3
Course instructor: Dr. Olena Syrotkina

String, Input/Output. Introduction to Lists.

This content is protected and may not be shared, uploaded, or distributed. Any unauthorized distribution, reproduction or sharing of this material, including uploading to CourseHero, Chegg, StuDocu or other websites, is strictly prohibited and may be subject to legal action. Students are granted access to this material solely for their personal use and may not use it for any other purpose without the express written consent of the instructor.

OBJECTIVES:

- i. To explore *str* and *list* data types and their built-in methods.
- ii. To use indices, slices, and concatenation of *strings* and *lists*.
- iii. To accept user input entered through the keyboard.

1. Launch **IDLE** in your computer, open a **new file** through IDLE, and name it **lab3sol.py**.

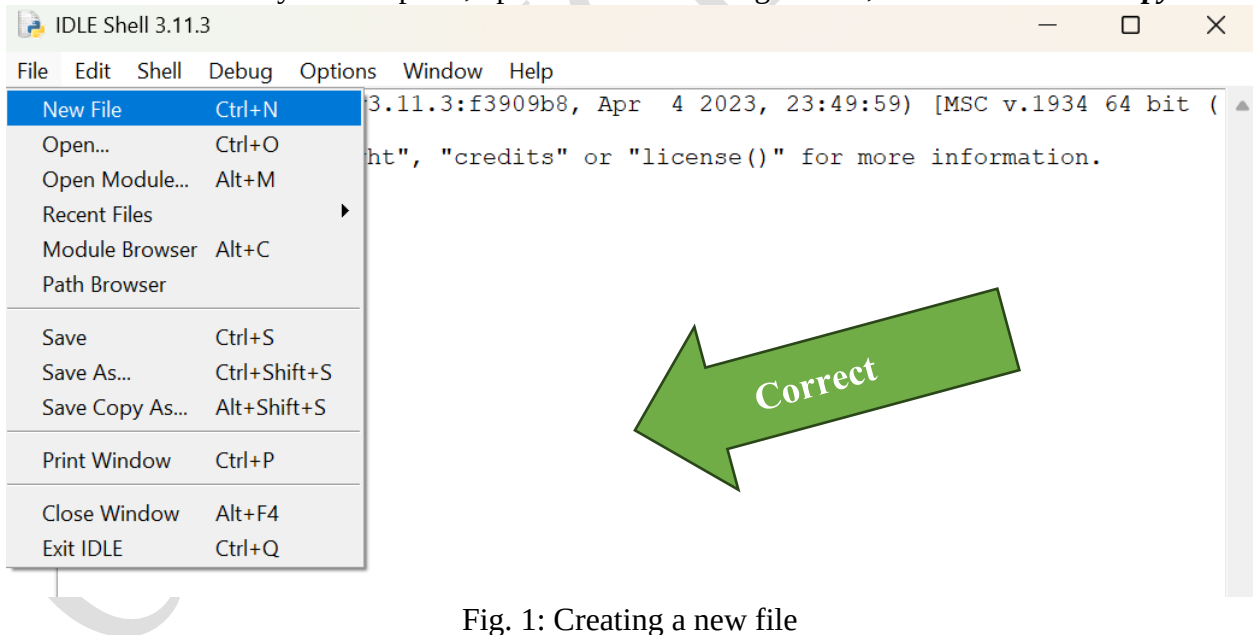


Fig. 1: Creating a new file

2. In *lab3sol.py*, write a Python program to perform the following tasks:
 - a. Ask the user to input the name of their **favorite Canadian province or territory** in lowercase letters and assign it to a variable **fav_province**. Then, print it using uppercase letters with a *suitable message*¹.

- b. Create a new string where the first (leftmost) occurrence of the letter 'e' in **fav_province** is replaced with '3' (**three**) and assign this new string to a variable **new_province_str**. Print **new_province_str** with a *suitable message*¹.
 - c. Ask the user to enter the year they first visited this province (between 1980 and 2025) and save this value as an integer variable **visit_year**. You may assume the user will enter a valid input. Print **visit_year** with a *suitable message*¹.
 - d. Define a list **geoList** = ['Toronto', 'Vancouver', 'Calgary', 'Quebec City', 'Ottawa', 'Halifax', 1867, ['Niagara Falls', 'CN Tower']] and print it.
 - e. Print the last 2 characters of the element of **geoList** with index 2 (which corresponds to a Canadian city).
 - f. Print all the elements of **geoList**, starting with the element at index 3 until the end of the list.
 - g. Ask the user to input the name of **another Canadian city, province, or landmark** they love and insert it at the beginning of **geoList**. Print the updated value of **geoList** with a *suitable message*¹.
 - h. Subtract the number of elements in **geoList** from **visit_year** and store the result in a variable called **result**. Print the number of elements in **geoList** and **result** with a *suitable message*¹.
3. **Submit your Python code** (i.e., as lab3sol.py file) through **Brightspace**.

¹**NOTE:** In this course, the phrase 'suitable message' refers to some kind of description that describes what is being printed. For example, just printing the value 22 by itself does not tell the user anything useful; however, printing 'x = 22' would add some context to what is being printed and so this would qualify as a 'suitable message' if the value of x is to be printed.

²**NOTE:** **Do NOT submit a screenshot** or the contents of the IDLE shell displaying your output. **These will NOT be marked.** You must submit the *lab3sol.py* file.

Below is a **sample** screenshot showing example outputs for Lab 3.

Please note: **do not submit a screenshot; only submit your lab3sol.py file.** Your suitable message does not need to match the screenshot exactly.

```

IDLE Shell 3.11.3
File Edit Shell Debug Options Window Help
Python 3.11.3 (tags/v3.11.3:f3909b8, Apr 4 2023, 23:49:59) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/OS/U of W/=====Summer 2025=====2067/Labs/Solutions/Summer2025/lab3sol.py
Enter the name of your favorite Canadian province or territory in lowercase letters: quebec
Your favorite province or territory in uppercase is: QUEBEC
After replacing the first 'e' with '3', the new string is: qu3bec
Enter the year you first visited this province or territory (between 1980 and 2025): 2021
You first visited this province or territory in: 2021
Here is the original geoList: ['Toronto', 'Vancouver', 'Calgary', 'Quebec City', 'Ottawa', 'Halifax', 1867, ['Niagara Falls', 'CN Tower']]
The last 2 characters of the city at index 2 ('Calgary') are: ry
Elements of geoList from index 3 to the end: ['Quebec City', 'Ottawa', 'Halifax', 1867, ['Niagara Falls', 'CN Tower']]
Enter the name of another Canadian city, province, or landmark you love: Montreal
Updated geoList after inserting your new favorite: ['Montreal', 'Toronto', 'Vancouver', 'Calgary', 'Quebec City', 'Ottawa', 'Halifax', 1867, ['Niagara Falls', 'CN Tower']]
The number of elements in geoList is: 9
Result of subtracting the number of elements from your visit year: 2012
>>>

```

Fig. 2. A sample screenshot showing example outputs for Lab 3.

Submission:

1. You will earn a maximum of 100 points (accounts for 2.5%) for submitting your **source code** (*lab3sol.py*) within the specified deadline.
2. **Do not upload your source code to Brightspace as a single zip/rar file.** Such submissions will not be accepted and a mark of **zero** will be assigned.
3. Marks will be **deducted** if a ***suitable message and/or comments*** are missing.
4. Late submissions will incur a penalty of 10% per day for **up to two days**. Submissions made beyond two days will not be accepted, and a mark of **zero** will be assigned.
5. Unlimited resubmissions are allowed. But keep in mind that we will consider the last submission. That means that if you resubmit after the deadline, a penalty will be applied, even if you submitted an earlier version on time.
6. Your grace period is **5 hours**.

Academic Integrity.

Plagiarism is a serious offense and can result in severe consequences such as receiving a grade of **zero** on the assignment/lab or even being asked to leave the program.

Copying or using someone else's code is considered plagiarism. This includes using code from online sources, previous labs/assignments, or from other students. Even if you have modified the code, our antiplagiarism software can still show it as plagiarism.

To avoid plagiarism, make sure to always use your own words and ideas when writing source code. Additionally, always check with your instructor to make sure you understand what is allowed and what is not in terms of using outside resources.

Remember that academic integrity is essential for your own personal and professional growth, and it is your responsibility to uphold these principles. So please take it seriously and always produce your own original work.